

C# Console Application with Login & User Creation (DAL + Service)

1. Model Layer – User.cs

```
namespace Models
{
    public class User
    {
        public string Username { get; set; }
        public string Password { get; set; }
    }
}
```

2. DAL Layer – UserDAL.cs

```
using System.Data.SqlClient;
using Models;

namespace DAL
{
    public class UserDAL
    {
        private string connectionString =
            "Server=localhost;Database=AuthDB;Trusted_Connection=True;";

        public bool ValidateUser(User user)
        {
            using (SqlConnection con = new SqlConnection(connectionString))
            {
                string query = "SELECT COUNT(*) FROM Users WHERE Username=@u AND Password=@p";
                SqlCommand cmd = new SqlCommand(query, con);

                cmd.Parameters.AddWithValue("@u", user.Username);
                cmd.Parameters.AddWithValue("@p", user.Password);

                con.Open();
                int count = (int)cmd.ExecuteScalar();
                return count == 1;
            }
        }

        public bool AddUser(User user)
        {
            using (SqlConnection con = new SqlConnection(connectionString))
            {
                string query = "INSERT INTO Users (Username, Password) VALUES (@u,@p)";
                SqlCommand cmd = new SqlCommand(query, con);

                cmd.Parameters.AddWithValue("@u", user.Username);
                cmd.Parameters.AddWithValue("@p", user.Password);

                try
                {
                    con.Open();
                    cmd.ExecuteNonQuery();
                    return true;
                }
                catch
                {
                    return false;
                }
            }
        }
    }
}
```

3. Service Layer – UserService.cs

```
using DAL;
using Models;

namespace Services
{
    public class UserService
    {
        UserDAL userDAL = new UserDAL();

        public bool Login(string username, string password)
        {
            User user = new User
            {
                Username = username,
                Password = password
            };
            return userDAL.ValidateUser(user);
        }

        public bool Register(string username, string password)
        {
            User user = new User
            {
                Username = username,
                Password = password
            };
            return userDAL.AddUser(user);
        }
    }
}
```

4. UI Layer – Program.cs

```
using System;
using Services;

class Program
{
    static UserService userService = new UserService();
    static bool isLoggedIn = false;

    static void Main()
    {
        while (true)
        {
            Console.Clear();
            Console.WriteLine("1. Login");
            Console.WriteLine("2. Create User");
            Console.WriteLine("3. Exit");
            Console.Write("Choice: ");

            int choice;
            int.TryParse(Console.ReadLine(), out choice);

            switch (choice)
            {
                case 1:
                    Login();
                    break;

                case 2:
                    if (isLoggedIn)
                        CreateUser();
                    else
                        ShowError("Please login first.");
                    Pause();
                    break;

                case 3:
                    return;

                default:
                    ShowError("Invalid choice.");
                    Pause();
                    break;
            }
        }
    }

    static void Login()
    {
        Console.Write("Username: ");
        string username = Console.ReadLine();

        Console.Write("Password: ");
        string password = Console.ReadLine();

        if (userService.Login(username, password))
```

```

        {
            isLoggedIn = true;
            Console.WriteLine("Welcome " + username);
        }
        else
        {
            ShowError("Incorrect username or password.");
        }
        Pause();
    }

    static void CreateUser()
    {
        Console.Write("New Username: ");
        string username = Console.ReadLine();

        Console.Write("New Password: ");
        string password = Console.ReadLine();

        if (userService.Register(username, password))
            Console.WriteLine("User created successfully.");
        else
            ShowError("Username already exists.");
    }

    static void ShowError(string msg)
    {
        Console.ForegroundColor = ConsoleColor.Red;
        Console.WriteLine(msg);
        Console.ResetColor();
    }

    static void Pause()
    {
        Console.WriteLine("Press any key...");
        Console.ReadKey();
    }
}

```